CAMLIN TAN

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EDUCATION

National University of Singapore **Aug 2024 - Dec 2027 (Expected)**
Bachelor of Computing in Computer Science

- GPA : 4.88 / 5.0
- Awards: Dean's List 24/25
- Availability: Only available for roles starting January 2027 onwards (due to scheduled overseas university exchange).
- Relevant course work: Data Structures & Algorithms, Software Engineering, Operating Systems, Object-Oriented Programming, AI and Machine Learning, Introduction to Information Security

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, TypeScript, JavaScript, HTML/CSS.
Backend & Systems: FastAPI, Node.js, Server-Sent Events (SSE), WebSockets, RESTful APIs.
Data & AI Infrastructure: SQLite, Vector Databases (Chroma, Supabase), LangGraph, LlamaIndex, RAG Pipelines.
DevOps & Tools: Docker, Git/GitHub, GitHub Actions (CI/CD), JUnit, Vercel.
Frontend Frameworks: React.js, Tailwind CSS.

WORK EXPERIENCE

HTX (Home Team Science and Technology Agency) **Singapore**
Software Engineering Intern (Backend & AI Infrastructure) **May 2026 - Jul 2026**

- Engineered a checkpointed, human-in-the-loop orchestration engine for an agentic software developer harness using Python, FastAPI, LangGraph, and SQLite, coordinating requirements, planning, code generation, validation, and repair across a 9-node state graph.
- Implemented a dependency-aware generation engine utilizing Depth-First Search (DFS) for topological sorting and cycle validation to guarantee execution order.
- Built an isolated execution sandbox utilizing the Docker SDK to automate build validations, handle asynchronous container logs, and manage thread-safe process cancellation.
- Architected real-time SSE streams and a WebSocket proxy, implementing atomic temporary file replacements and node-level rollback to mitigate partial-write failures.
- Authored asynchronous integration tests utilizing Python testing libraries to validate context isolation and patch safety.

Teaching Assistant, National University Singapore, Singapore **Aug 2025 - Nov 2025**

- Facilitated learning for 20 undergraduates in CS1231S (Discrete Structures), translating complex abstract concepts (Logic, Graph Theory) into actionable understanding.

Part-time Tutor, A-Levels Tutor, Singapore **Dec 2023 - Jul 2024**

- Conducted personalised coaching in Mathematics and Sciences for A-Level students, resulting in an average grade improvement of 2 to 3 grades.

PROJECTS

Developer, Real-Time Quant Data Engine
(Python, PostgreSQL, TimescaleDB, Redpanda, FastAPI, PySpark, Airflow, Docker) **Jul 2026 - Present**

- Engineered an end-to-end financial data pipeline using Python and Docker, streaming live WebSocket market data into a Redpanda message broker.
- Processed continuous tick data into 60-second OHLCV bars using Pandas, storing the aggregated time-series data in PostgreSQL.
- Optimized high-volume data retrieval by migrating PostgreSQL tables to TimescaleDB hypertables, serving historical market data through an asynchronous FastAPI service.
- Orchestrated daily batch workflows using Apache Airflow and PySpark to extract, compress, and archive database records into MinIO (AWS S3 API) object storage as Parquet files.
- Ensured pipeline reliability by containerizing the multi-node system with Docker Compose and automating testing via GitHub Actions and pytest.

Developer, Multimodal RAG Retrieval System (Python, FastAPI, React.js, LlamaIndex) **Dec 2025 - Present**

- Architected a full-stack Retrieval-Augmented Generation (RAG) application, integrating a FastAPI backend with a responsive React frontend to serve real-time AI responses.
- Built a context-aware RAG pipeline using LlamaIndex to ground LLM responses in user-uploaded data, reducing model hallucinations on domain-specific queries through prompt engineering and context injection.
- Engineered a multimodal transformation workflow converting unstructured image nodes into semantic text descriptions, enabling hybrid search across both text and visual data.
- Collaborated in a 2-person team, pair-programmed core modules and used GitHub PRs/code reviews to ship features faster with fewer defects.

Developer, HealthNote (Java, JUnit, CI/CD) **Sep 2025 - Nov 2025**

- Ranked Top 10% in Cohort for software engineering excellence, delivering a production-grade system compliant with strict OOP principles.
- Worked in a 5-person Agile team, reviewed PRs, and coordinated test ownership.
- Designed and implemented a robust CI/CD pipeline using GitHub Actions, automating testing and build workflows to ensure code stability.
- Wrote comprehensive automated tests using JUnit to ensure code coverage and prevent regression in a modular architecture.
- Enforced code maintainability and documentation standards, facilitating easy collaboration and future feature scaling.

Developer, TravelSync (React.js, Supabase, Google Maps API) **May 2025 - Aug 2025**

- Collaborated in a team of 2 to engineer a real-time collaborative travel planner using React and Supabase, synchronising state across multiple concurrent client sessions.
- Integrated Google Maps API to render dynamic routes and location markers, optimising the user experience for complex itinerary visualisation.